

Project Title: Student Grade Predictor

Name: Roshan Kushwaha(9837)

Internship Program: Summer Internship

University: KIIT University

Department: Computer Science and Engineering

Submission Date: *(17th june 2026)*

Abstract

The Student Grade Predictor is a machine learning project designed to estimate a student's final academic performance using various input parameters such as attendance percentage, assignment scores, midterm marks, and daily study hours. The objective of this project is to demonstrate how predictive analytics can assist educational institutions in identifying students who may require additional support.

A synthetic dataset containing student performance records was generated and processed using Python libraries including Pandas and NumPy. A Linear Regression model was trained using Scikit-learn to learn the relationship between the input variables and the final grade. The dataset was divided into training and testing sets to evaluate the model's performance.

The model was assessed using Mean Absolute Error (MAE) and R^2 Score metrics. Visualization techniques using Matplotlib were employed to compare actual and predicted grades, providing clear insights into model accuracy.

This project demonstrates the practical application of machine learning in education and highlights the potential of data-driven approaches to improve academic outcomes.

Introduction

In recent years, machine learning has become an important tool for solving real-world problems across various domains, including education. Educational institutions generate large amounts of data related to student performance, attendance, assignments, and examinations. Analyzing this data can help educators identify patterns and make informed decisions to improve academic outcomes.

The Student Grade Predictor project aims to use machine learning techniques to estimate a student's final grade based on factors such as attendance percentage, assignment scores, midterm examination marks, and study hours. By predicting academic performance in advance, teachers and institutions can identify students who may need additional support and guidance.

This project utilizes the Linear Regression algorithm, a supervised machine learning technique used for predicting continuous numerical values. The model is trained using a dataset containing student performance records and is evaluated using performance metrics such as Mean Absolute Error (MAE) and R^2 Score.

The project was developed using Python and popular data science libraries including Pandas, NumPy, Scikit-learn, and Matplotlib in the Google Colab environment. The results demonstrate how machine learning can be applied effectively in the education sector to support data-driven decision-making.

Objectives

1. Predict student grades using machine learning.
2. Analyze factors affecting academic performance.
3. Visualize prediction accuracy.

Technologies Used

1. Python
2. Google Colab
3. Pandas
4. NumPy
5. Scikit-learn
6. Matplotlib
7. GitHub

Dataset Description

The dataset used in this project was synthetically generated using Python libraries such as Pandas and NumPy. The dataset contains information related to student academic performance and consists of 100 records. Each record represents an individual student and includes various factors that may influence the final grade.

The dataset contains the following attributes:

- **Attendance:** Represents the student's attendance percentage, ranging from 60% to 100%.
- **Assignment_Score:** Represents the marks obtained in assignments, ranging from 50 to 100.
- **Midterm_Score:** Represents the marks obtained in the midterm examination, ranging from 45 to 100.
- **Study_Hours:** Represents the average number of hours a student studies per day, ranging from 1 to 7 hours.
- **Final_Grade:** Represents the predicted final grade of the student, calculated using a weighted formula based on the input attributes.

The dataset was saved in CSV format as `student_grades.csv` and used to train and evaluate the machine learning model.

Methodology

The Student Grade Predictor project follows a structured machine learning workflow to predict students' final grades.

1. **Data Generation:** A synthetic dataset containing 100 student records was created using Python. The dataset includes attendance percentage, assignment scores, midterm scores, study hours, and final grades.
2. **Data Preprocessing:** The generated dataset was loaded into a Pandas DataFrame and checked for consistency. Since the dataset was synthetically generated, it contained no missing values or duplicate records.
3. **Feature Selection:** Four independent variables—Attendance, Assignment_Score, Midterm_Score, and Study_Hours—were selected as input features. The Final_Grade column was selected as the target variable.
4. **Data Splitting:** The dataset was divided into training and testing sets using an 80:20 ratio. The training data was used to train the model, while the testing data was used to evaluate its performance.
5. **Model Training:** A Linear Regression algorithm from the Scikit-learn library was used to train the prediction model.
6. **Model Evaluation:** The model's performance was assessed using Mean Absolute Error (MAE) and R^2 Score metrics.
7. **Prediction and Visualization:** The trained model was used to predict the final grade of a new student based on input values. A scatter plot of actual versus predicted grades was generated using Matplotlib to visualize the model's accuracy.

Results and Screenshots

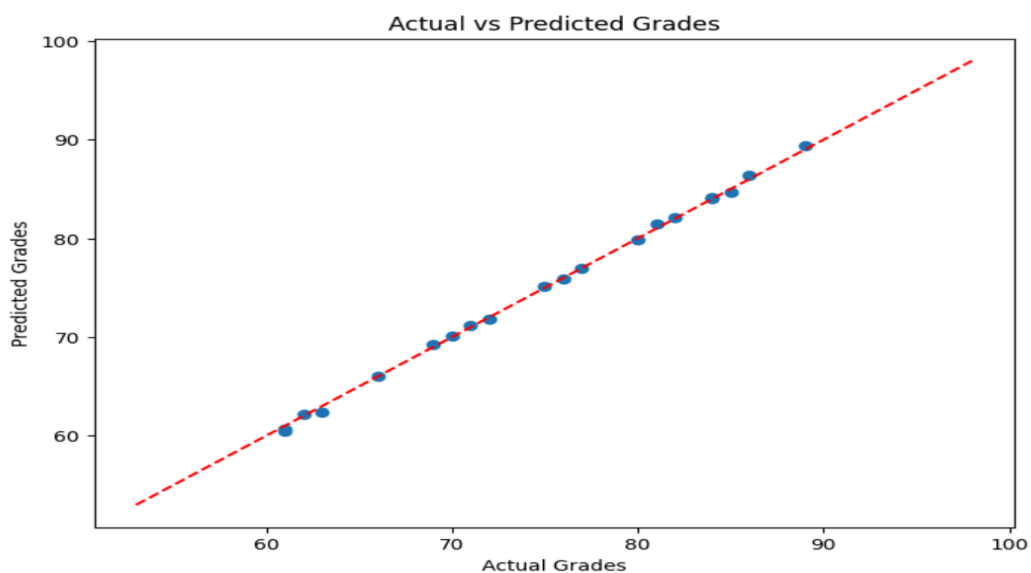
1. Dataset preview (`df.head()`)
2. MAE and R^2 score output
3. Predicted grade output
4. Scatter plot

...

	Attendance	Assignment_Score	Midterm_Score	Study_Hours	Final_Grade
0	98	97	45	6	81.0
1	88	72	63	3	72.0
2	74	73	46	5	66.0
3	67	86	97	1	78.0
4	80	84	88	4	84.0

Mean Absolute Error: 0.22
 R^2 Score: 1.0

Predicted Final Grade: 88.86
/usr/local/lib/python3.12/dist-packages
warnings.warn(



Conclusion

Mention that the model successfully predicted student grades with high accuracy and demonstrated the usefulness of machine learning in education.

References

1. Python Documentation
2. Scikit-learn Documentation
3. Pandas Documentation
4. Matplotlib Documentation